Amit Singh

SENIOR SOFTWARE ENGINEER · SDE-3

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SUMMARY

Senior Software Engineer with 7 years building distributed backend systems in Go and Java / Spring Boot at Tekenlight Solutions, across FinTech, banking and multi-tenant Ed-Tech. 5+ years of production Go on a mobile banking platform: built shared security infrastructure — ECDSA/RSA payload signing with a centralized gRPC key registry, JWT (RS512) and multi-provider OAuth2 verification, PII-safe audit logging — and led a zero-downtime Visa-to-Mastercard network migration. In Java, architected an ed-tech platform serving 10K+ daily active users, with a data-driven rule engine for per-school workflows, AI-backed semantic search and a self-hosted OTA update service. Leads engineering teams of 10 and 8 across the two products.

TECHNICAL SKILLS

Languages	Java, Go / Golang, Python, Dart (Flutter), SQL
Frameworks	Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, gRPC, FastAPI
Data Stores	PostgreSQL, Redis, HBase, Hive, Elasticsearch, Milvus
Data Pipelines	Hadoop, Apache Airflow
AI / LLM	MCP (Model Context Protocol), Claude, OpenAI, Ollama, LiteLLM, RAG pipelines, prompt engineering, tool orchestration
Cloud & DevOps	Azure, Firebase, Docker, CI/CD pipelines, Git
Architecture	Microservices, Event-Driven Architecture, Distributed Systems, Rule Engine & Workflow Configuration, Multi-Tenant SaaS, Modular / Layered Design, Distributed Caching, Real-Time Data Processing
Security	ECDSA (P-256/SHA-256) & RSA (PKCS1v15) signing, PEM/PKCS8 key parsing, centralized key registry (gRPC), JWT (RS512), OAuth2, SSO, RBAC, field-level PII encryption

EXPERIENCE

Tekenlight Solutions Pvt Ltd

Bengaluru, India · 2019 — Present

Senior Software Engineer 2023 — PRESENT **Software Engineer** 2019 — 2023

Ed-Tech Web Application

Java, Spring Boot, PostgreSQL, Milvus, Python · 2024 — Present

- Architected and led backend development of an AI-enabled, multi-tenant ed-tech platform on Java, Spring Boot and PostgreSQL, serving 10K+ daily active users with high-volume, real-time data flows across teachers, school administrators, students and parents.
- Designed a configurable, data-driven rule engine that lets each school define its own workflows — approvals, escalations, notifications, grading and eligibility policies — without code changes, turning a single-tenant application into an extensible multi-tenant platform.
- Built AI-backed semantic content search that lets teachers and students find learning material by meaning rather than keywords — embeddings indexed in Milvus, served by a Python FastAPI service with LiteLLM as the model-provider layer, alongside Elasticsearch.
- Built a self-hosted mobile over-the-air (OTA) update service in Java implementing the Expo Updates v1 protocol end to end — signed multipart manifests, RSA code signing via Azure Key Vault, remote rollback, Redis-cached delivery — replacing a paid third-party service and letting the team ship new features the same day, bypassing App Store and Google Play review cycles.
- Designed a modular backend architecture on reusable frameworks, raising release frequency from biweekly to weekly — including an annotation-driven generic CRUD framework that removed ~40% of boilerplate service code across 15+ modules and standardized data access.
- Engineered a pluggable multi-layer caching framework (in-memory + Redis) that cut average database read latency ~45% under peak load.
- Made the platform PII-compliant with application-layer field-level encryption — key rotation and transparent encrypt/decrypt at the persistence boundary — and secured access with SSO and fine-grained RBAC on Spring Security.
- Led a cross-functional team of 10 engineers — sprint planning, code reviews, mentorship — and cut new-hire onboarding time by ~30%.

FinTech Mobile Banking Application

Golang, gRPC, Flutter · 2021 — Present

- Led the business-driven migration of the card payment backend from Visa to Mastercard — card issuance, transaction routing, settlement flows and compliance hooks — with zero downtime and no customer-facing disruption.
- Architected the Golang microservices backend with custom authentication, structured logging and connection pooling, improving throughput ~35% over the legacy monolithic service.
- Built a shared Go cryptographic signing library adopted by the Ledger, KYC, RTP and RTAC modules — ECDSA (P-256/SHA-256) and RSA (PKCS1v15) payload signing, PEM/PKCS8 key parsing, and a centralized gRPC key registry with per-domain key isolation and a secure file-based fallback.
- Engineered JWT (RS512) issuance and multi-provider OAuth2 verification middleware supporting Firebase, Azure AD, custom OAuth and anonymous sessions, unifying authentication across all platform services.

- Designed a Go audit-logging middleware with correlation-ID request tracing, PII-safe field sanitization and complete request/response audit trails to meet compliance requirements.
- Delivered debit card applications, fund transfers and Apple Pay / Google Pay integration in the Flutter mobile app, handling high-frequency transaction data in real time.
- Led an 8-person backend and mobile team, introducing modular framework patterns that reduced technical debt and enabled parallel feature development; critical services sustained a 99.9% uptime SLA.

Banking Solutions — Financial Reporting System Java, Spring Boot, Hadoop, HBase, Airflow · 2019 — 2021

- Developed Java Spring Boot REST APIs for automated financial report generation in BAI2, CAMT052 and CAMT053 formats for a banking client's compliance reporting pipeline.
- Orchestrated data ingestion from the bank's data lake into Hadoop with Apache Airflow, enabling reliable daily batch processing of machine- and human-readable reports.
- Selected HBase over Hive for random-access report reads, reducing p99 query latency from ~800 ms to ~120 ms at 2M+ daily rows; containerized services with Docker for scalable deployment.

EDUCATION

B.Tech, Mechanical Engineering — Raj Kumar Goel Institute of Technology (A.K.T.U) 2017